

Cybersecurity in Healthcare: Identifying Threats, Impacts, and Practical Solutions

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Northwestern State University students in the Computer Information Systems program are conducting research on cybersecurity threats facing medical facilities as part of the 2025-2026 cybersecurity grant coordinated with Natchitoches Regional Medical Center (NRMC). This paper examines major cybersecurity threats to medical facilities, with focus on their definitions and impacts. The goal is to provide actionable insights that NRMC and similar facilities can use to strengthen defenses while minimizing disruption to daily operations. Research comes from industry reports, government data, and recent incidents that highlight the urgency of proactive measures in an era of increasing digitization of healthcare data. This work aims to connect academic research with practical applications for community partners like NRMC.

Major cybersecurity threats and attack types at medical facilities include phishing, ransomware, malware, supply chain attacks, insider threats, and exploited vulnerabilities. Phishing involves fake emails, texts, or messages that appear legitimate and create a false sense of urgency to trick users into clicking malicious links or revealing login credentials.

Ransomware is malicious software that encrypts computers and patient records, rendering them unusable until a ransom is paid, which can halt access to lab results or billing systems. Malware refers to any harmful software secretly installed to steal data, damage files, or create back doors for attackers. Supply chain attacks occur when hackers breach a trusted vendor, third-party system, or software tool used by the hospital to gain access to the hospital's network. Insider threats are from individuals inside the organization who, either accidentally or intentionally, click malicious links or misuse their access. Exploited vulnerabilities, often through unpatched systems, allow attackers to take advantage of outdated software or known security holes, creating an entry point into restricted systems. These threats rarely work in isolation—they often combine into multi-layered attacks that are harder to detect and stop.

Medical facilities like NRMHC face cybersecurity threats that attempt to compromise digital systems, networks, or data and can result in unauthorized access, disruption of services, or theft of sensitive information. These threats mean more for medical facilities who are trusted with protected health information such as patient records, medical histories, and billing details. According to Dialog Health (2025), 92% of healthcare organizations were targeted by cyber attacks in the past 12 months, with 90% experiencing at least one breach. Hacking and IT incidents account for over 80% of large breaches, while unauthorized access and disclosure incidents accounted for approximately 20% of data breaches (HIPAA Journal, 2026a). According to Dialog Health (2025), the healthcare sector had an expected investment of \$125 billion in cybersecurity tools and services from 2020-2025, showing a 15% annual growth rate; however, vulnerabilities are still present largely due to outdated legacy operating systems.

According to Dialog Health (2025), in 2025 the average cost per breached record in healthcare was \$408, three times the overall industry average of \$148, outlining the

comparatively large financial and operational burden this has on the healthcare system. The U.S. Department of Health and Human Services Office for Civil Rights reported 7,419 large breaches since 2009, with 710 in 2025 alone affecting around 62 million people (HIPAA Journal, 2026a). These threats not only endanger patient privacy but can cause critical interference with patient care that could lead to delayed treatments or worse.

Globally and in the U.S., cybersecurity threats to medical facilities are being reported more often, with a 21% increase in attacks in 2025 (Bruce, 2026). According to the Verizon Data Breach Investigations Report (2025), there were 1,710 security incidents in healthcare with 1,542 confirmed data disclosures, primarily from system intrusions like ransomware and espionage. Dialog Health (2025) notes that common threats include ransomware, phishing, and supply chain vulnerabilities, and that ransomware caused an average of nearly 19 days of downtime.

The most impactful threats are ransomware and hacking. According to Dialog Health (2025), ransomware attacks doubled between 2016-2021 and often exploit vulnerabilities like outdated systems or weak credentials. The largest breach on record, the 2024 Change Healthcare ransomware attack, affected 192.7 million people worldwide, causing disruptions in billing and care (HIPAA Journal, 2026a). Emsisoft (2025) reported that 85 U.S. healthcare networks (1,031 U.S. hospitals) were hit by ransomware attacks throughout 2024.

One of the most costly events, the 2025 Conduent Business Services breach, affected over 25 million people, exposing names, SSNs, and treatment information, resulting in reported direct costs of \$25 million (HIPAA Journal, 2026b). DaVita's April 2025 ransomware attack impacted 2.69 million, involving network encryption and data exfiltration, disrupting dialysis services but not critical care—costs have been estimated to be in the millions due to recovery and notifications (HIPAA Journal, 2026b). The Yale New Haven Health System breach in March

2025 affected 5.56 million, with hackers stealing demographic and SSN data for weeks before being detected (HIPAA Journal, 2026b). These incidents were possible by exploiting vulnerabilities such as unpatched software and phishing, leading to average healthcare breach cost of \$7.42 million (HIPAA Journal, 2026). Data from the HHS Office for Civil Rights confirms 710 large healthcare data breaches occurred in 2025, demonstrating why proactive security is essential (HIPAA Journal, 2026a).

In our own part of the country, the threats look a lot like the national picture but seem more impactful because they are closer to home. Smaller hospitals in our region tend to run on older systems with tighter budgets, which makes ransomware and phishing the most common and impactful attacks. According to HIPAA Journal (2026a), states across the South like Georgia, Texas, and Louisiana continue to see a steady stream of breaches, with hacking and IT incidents making up the majority of cases.

Two of the most impactful events in the southern region happened in the last 2 years in Georgia. The first was the June 2025 Aflac breach. Hackers gained unauthorized access through suspicious activity on a limited number of the company's U.S. systems. The attack was a classic hacking incident that took advantage of what appears to have been a social engineering opening. Aflac spotted it quickly on June 12, contained the whole thing within hours by working with third-party cybersecurity experts and notifying federal law enforcement, secured the affected accounts, reset passwords, and stepped up monitoring. Even so, the breach ended up exposing names, SSNs, health information, and other personal data for roughly 13.9 million people in the U.S. (Aflac Incorporated, 2025; HIPAA Journal, 2026b).

The second was the November 2024 Lockton Southeast breach (reported in early 2025). This was also a hacking/IT incident in which an unauthorized party got into a single computer

and user account inside the company's network. The weak spot was a lack of strong controls around that individual account. Lockton caught the suspicious activity right away, brought in third-party cybersecurity experts, notified law enforcement, and did a full review of the files. They then rolled out additional security measures across their systems to prevent it from happening again. The breach still ended up affecting more than 1.1 million people whose personal health data was in those files (Southeast Series of Lockton Companies, LLC, 2025; HIPAA Journal, 2026b).

Two of the most impactful events in Louisiana show exactly why facilities like NRMC need to be aware of threats and how to fight against them. The first was the Lafourche Medical Group phishing attack in 2021. Hackers tricked someone into clicking a phishing email and gained access to an owner's email account that contained protected health information. The breach exposed medical records, diagnoses, lab test results, and other protected health information for 34,862 patients. Lafourche Medical Group could not identify exactly which records were viewed, so they notified every patient. The U.S. Department of Health and Human Services Office for Civil Rights investigated and found the group had never conducted a proper risk analysis or regularly reviewed systems activity logs. The case settled for \$480,000 in December 2023. This was the first HIPAA settlement the Office of Civil Rights has ever resolved specifically for a phishing attack (HIPAA Journal, 2023; U.S. Department of Health and Human Services, 2023).

The second was the Lake Charles Memorial Health System ransomware attack in 2022. An unauthorized party, later claimed by the Hive ransomware group, gained access to the network and exfiltrated patient data affecting 269,752 individuals across multiple locations and clinics. The exposed information included names, addresses, dates of birth, Social Security

numbers, medical record numbers, health insurance details, payment information, and limited clinical information. A related class-action lawsuit later reached a \$2 million settlement in 2025 (HIPAA Journal, 2022; HIPAA Journal, 2025).

Both incidents in Louisiana were enabled by the same latent organizational weakness we see across smaller facilities in central Louisiana such as tight budgets, older systems that are hard to keep updated and patched, limited staff for security monitoring, and gaps in basic controls like risk analysis and vendor oversight. These are exactly the kinds of everyday vulnerabilities that make phishing and ransomware so successful in our part of the state.

What makes medical facilities in central Louisiana and the southern U.S. region most vulnerable is a mix of tight budgets, older technology, and not enough staff to stay on top of security. According to Dialog Health (2025), many facilities still rely on legacy systems that are hard to patch, and human error plays a role in 31% of data loss incidents. The most common ways attacks succeed here are through phishing emails that trick busy employees into clicking bad links and by slipping in through unpatched systems or weaker vendor connections. These issues point to what experts call latent organizational weaknesses (LOWs). LOWs are underlying problems like insufficient training, outdated policies, and limited resources that create gaps attackers can easily exploit.

Medical facilities can strengthen their defenses by combining proven strategies with practical improvements that complement their current security plans. One effective proven strategy is to conduct thorough incident reports, hold LOW meetings, and maintain up-to-date incident response plans after any security event. These steps help teams identify root causes, improve procedures, and respond faster if an attack happens again. Regular policy reviews and

timely software updates are keys to a successful protection plan, as they close known gaps before attackers can use them.

In addition to these established practices, many organizations are adopting more proactive and innovative approaches. Regular cybersecurity training that uses realistic scenarios from daily operations has proven helpful in reducing human error, especially with phishing attempts (Guardz, 2025). Other practical steps include keeping software and systems updated on a regular basis, strengthening vendor security reviews, and implementing multi-factor authentication of critical systems where possible. These measures can add important layers of protection while fitting into existing operations at facilities like NRMCC.

Data from industry reports supports these strategies. According to Dialog Health (2025), organizations that provide regular cybersecurity awareness training and clear policies see lower rates of successful phishing and ransomware incidents. HIPAA Journal (2026a) also shows that facilities that actively address legacy system issues and improve basic security controls tend to experience fewer large-scale breaches over time. By building on current efforts with practical, targeted improvements, medical facilities can strengthen their defenses against the threats discussed while continuing to focus on high-quality patient care.

In summary, cybersecurity threats to medical facilities, including ransomware, phishing, and exploited vulnerabilities, pose significant risks both nationally and in the southern United States. By understanding these threats, their local impacts, and the underlying latent organizational weaknesses that enable them, facilities like NRMCC can implement practical, targeted improvements. Combining proven strategies such as incident reviews and LOW meetings with innovative approaches like realistic training and regular system updates provides a balanced path forward. These measures are achievable without adding undue burden to daily

operations and can help protect patient data while supporting high-quality care. Continued partnerships between our CIS program and local medical facilities like NRMC will make a real difference in strengthening cybersecurity across central Louisiana.

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